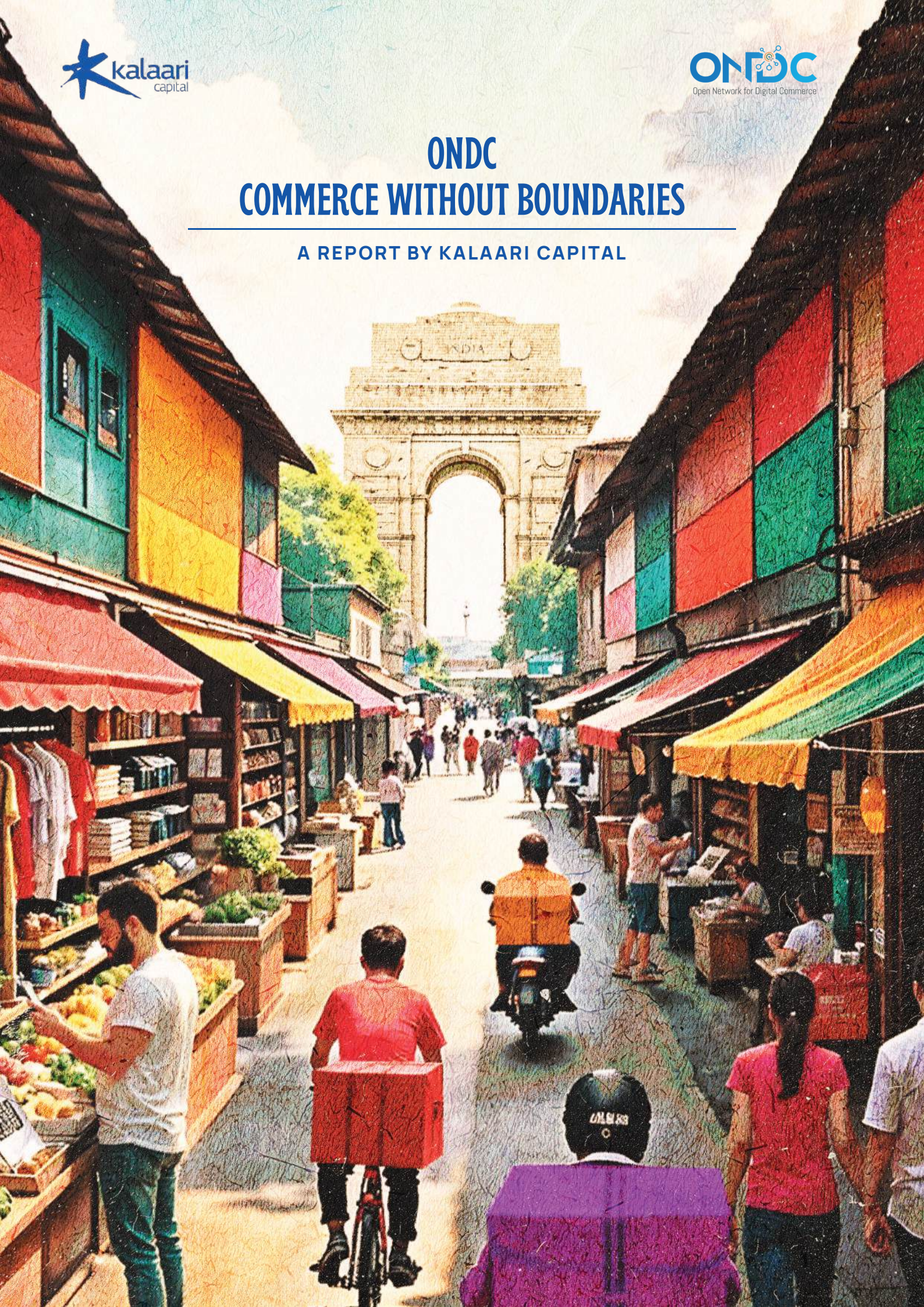
kalaari capital

ONDC
Open Network for Digital Commerce

ONDC
COMMERCE WITHOUT BOUNDARIES

A REPORT BY KALAARI CAPITAL



INTRODUCTION

In the heart of the world's largest democracy,
a digital revolution is unfolding at an unprecedented scale. India, with its 1.4 billion citizens, has emerged as a global digital powerhouse, transforming from a technology consumer to a technology creator in less than a decade.

The numbers tell a compelling story. From 954 million internet subscribers as of March 2024, to processing more than 15 billion digital transactions a month through UPI, India's digital footprint has expanded at a breathtaking pace. This transformation has helped the digital economy grow from contributing merely 4.5% of GDP in 2014 to a projected 20% by 2026.

The internet subscriber base now stands at

954.4 million,

with

556.05 million
in urban areas and

398.35 million
in rural area

India was home to

462.0 million
social media users in January 2024, equating to

32.2% of the total population

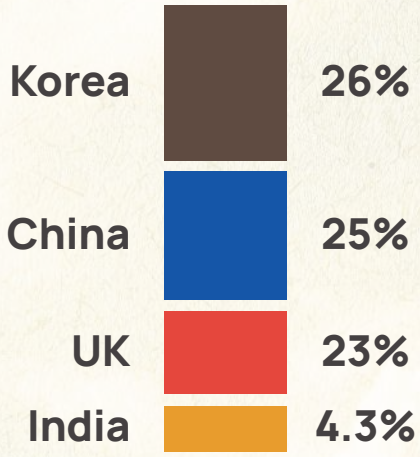
350 million
mature online users actively engaging in transactions

In a small shop in Banka, Bihar, a vegetable vendor completes her day's transaction using a QR code - one of the 15.4 billion digital payments processed in India just in Nov 2024. Meanwhile, in a tech hub in Bengaluru, a startup founder fine-tunes an AI Model that will serve millions. These contrasting yet complementary scenes paint a picture of India's digital revolution - a transformation so profound that it has become a case study in digital inclusion for the world.

Source: Investindia, Datareportal

© Copyright 2025 Kalaari Capital

Yet, these impressive figures mask an even more intriguing reality - we're just scratching the surface. While India boasts more than 900 million internet users, organized digital commerce reaches a fraction of them. Despite smartphones proliferating to the remotest villages, 80% of retail remains unorganised.



E-Retail Penetration

For every success story of a D2C brand scaling through e-commerce, there are thousands of local businesses still untouched by the digital revolution. This gap isn't just a challenge - it's perhaps the largest untapped digital opportunity globally.

We're looking at a potential digital economy of $1 trillion by 2028, but the real question is: How do we ensure this growth isn't just concentrated in our metropolitan centers but reaches the heart of Bharat?

Source: ONDC Website, Economic Times India

© Copyright 2025 Kalaari Capital

THE ANSWER LIES IN DIGITAL PUBLIC INFRASTRUCTURE (DPI)

DPI Is Not Just A Technological Advancement; It's A Socio-economic Game-changer For India.

DPI matters for India because it helps address systemic issues of access and equity. By providing an open, digital backbone, DPI reduces the cost of participation in the digital economy for small businesses, startups, and even individuals. For instance, Aadhaar has enabled millions to open bank accounts, while saving the country billions of dollars and helped millions of people access subsidies with ease.

The foundation of India's DPI initiatives rests on three fundamental pillars that ensure sustainable, inclusive growth:



India's digital transformation isn't limited to indentification or commerce. The country is witnessing a synchronized evolution of multiple Digital Public Infrastructure (DPI) initiatives that work in harmony to create a comprehensive digital ecosystem.

The National Health Stack (NHS) represents India's ambitious push to digitize healthcare delivery. Similar to how ONDC democratizes commerce, NHS aims to make healthcare more accessible and efficient. The impact is already visible: over 420 million digital health records have been created since its inception. This digital transformation is expected to reduce healthcare delivery costs by 20% while improving access to quality healthcare in rural area.

Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital

Agriculture, employing nearly half of India's workforce, is undergoing its own digital revolution through Agri Stack. This initiative aims to create a unified farmer database linked to land records, making it easier to deliver targeted benefits and services. The government plans to bring 6 crore farmers under this formal land registry system, potentially transforming how agricultural subsidies and services are delivered. The system will streamline the distribution of the ₹1.75 lakh-crore annual fertilizer subsidy, ensuring it reaches the intended beneficiaries.

Arguably one of the biggest example of this was in 2016, when India saw a digital revolution with the launch of the Unified Payments Interface (UPI), which transformed transaction processes and became a household name. By Dec 2023, UPI was handling over 12 billion transactions per month, valued at more than ₹18 trillion. UPI's architecture allowed it to break down the digital divide, extending the benefits of cashless transactions to the masses, including underserved sections of society like rural farmers, small vendors, and daily wage earners.

This penetration into smaller towns and rural areas has brought millions of previously unbanked individuals into the formal financial system. Matter of fact: UPI transactions at retail stores in rural and semi-urban India showed remarkable growth in 2023 with a 118% increase in transaction volume compared to 2022.

UPI's success lies in its simplicity. The ability to transfer money using just a mobile number or QR code has made it accessible even to those with limited technological literacy.

These DPI initiatives create powerful synergies and these platforms prove that the future of India’s digital economy lies not in closed, walled-off ecosystems, but in interoperable, scalable networks that invite participation from all.

Source: Business Standard

© Copyright 2025 Kalaari Capital

THE ECOMMERCE BOOM AND WHAT LIES AFTER

It began with urban consumers embracing the convenience of online shopping, with platforms like Flipkart and Amazon quickly gaining traction. With over 950 million internet users today and growing digital literacy, the e-commerce market hit significant milestones, reaching a **projected value of $325 billion by 2030.**

However, this growth, while impressive, has largely been dominated by a handful of players. **India's future economic growth isn't predicated on building monopolies** – a principle that lies at the core of ONDC's mission. These platforms have maintained control over the buyer-seller ecosystem, and while they provide efficiency, they also increase friction for sellers to dictate their own structures, fees, and guidelines.

AND YET - ALL OF INDIA'S ORGANISED COMMERCE STILL DOESN'T SERVE ALL OF INDIA

While organised retail and e-commerce have seen significant growth in India, they still face limitations in serving the entire country effectively.

Supply Side Problems

The Unorganized Sector's Dominance
Despite the rapid growth of e-commerce, the unorganized retail sector still accounts for over 84% of India's retail market. This sector, comprising millions of small, local shops, continues to be the primary source of goods for a large portion of the population, especially in non-metro areas.

Last-Mile Challenges
Organized commerce struggles with last-mile delivery in many parts of India. The complex geography, poor infrastructure in rural areas, and the high costs associated with reaching remote locations often make it economically unviable for large e-commerce players to service these areas effectively.

Source: Business Standard, Statista

© Copyright 2025 Kalaari Capital

Demand Side Problems

Cultural and Linguistic Diversity

India's vast cultural and linguistic diversity poses a significant challenge for organized commerce. With over 22 official languages and hundreds of dialects, creating a one-size-fits-all approach to commerce is nearly impossible. Local shops, on the other hand, are better equipped to cater to the specific needs and preferences of their communities.

Consumer Behavior And Trust

Traditional shopping habits run deep in India's consumer psyche. Local kirana stores operate on generations of trust and personal relationships, offering tailored services like:



Monthly credit arrangements



Personalized product recommendations



Home delivery based on a simple phone call



Product returns without formal processes

These deeply embedded consumer behaviours make it difficult for organised retail to replicate the same level of trust and personalisation at scale.

This disparity underscores a critical issue: The current e-commerce model is failing to penetrate deeper into India's heartland, leaving a vast untapped market and millions of potential consumers and sellers on the sidelines.

Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital

ENTER, OPEN NETWORK FOR DIGITAL COMMERCE

Open Network for Digital Commerce

By addressing these limitations and leveraging the power of DPI, ONDC aims to create a more inclusive digital commerce ecosystem that works for all of India, not just its urban centers. It seeks to bring the advantages of organized commerce to the unorganized sector, while preserving the local flavour and personal touch that characterises much of Indian commerce.

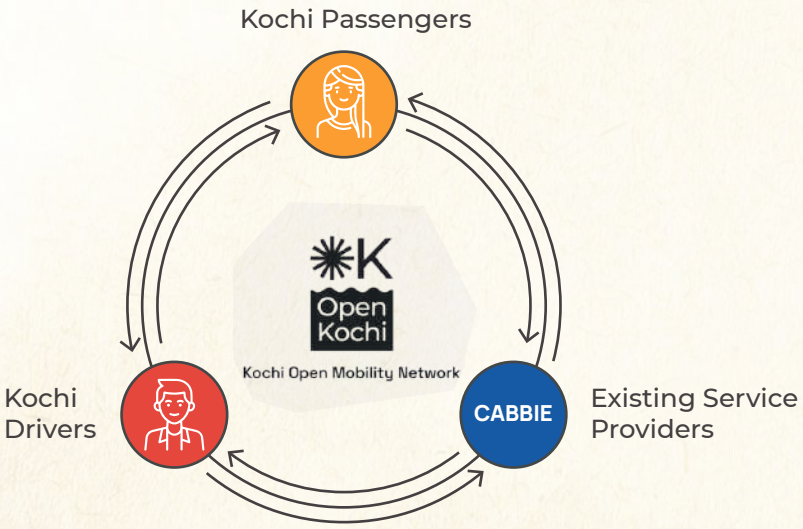
Where Did ONDC Come From?

In July of 2021, the Kochi Metropolitan Transport Authority (KMTA) launched what would become a pivotal experiment - the Kochi Open Mobility Network (KOMN), the world's first open mobility network.

The Kochi Experiment

KOMN demonstrated something remarkable - the power of an open protocol (Beckn) to transform how citizens accessed urban mobility services. The network allowed:

- Commuters to discover and book rides across multiple transport modes
- Auto and taxi drivers to operate without paying hefty commissions
- Local businesses to participate without being locked into specific platforms



Source: Openkochi

© Copyright 2025 Kalaari Capital

Within months, KOMN proved that an open network could create a more inclusive and efficient ecosystem. The success metrics were compelling:



From Mobility to Commerce

The success of KOMN caught the attention of policymakers and technologists who saw a broader application. If an open protocol could democratize mobility services, could the same approach transform digital commerce?This question led to the conceptualization of ONDC. The fundamental principles remained the same:

- Open protocols enabling interoperability
- Decentralized architecture
- Equal opportunity for all participants
- Focus on local ecosystem development

Source: Karan Kamble What Is Beckn Protocol

© Copyright 2025 Kalaari Capital

MORE IMPORTANTLY, WHAT IS ONDC?

First things first - it's neither a platform nor an app. It is an open network built on open-source protocols that allows buyers and sellers to transact directly, without being tied to any specific platform.

More simply put, ONDC is like an online shopping network where any seller, big or small, can connect with buyers, without having to register individually on each big platforms like Amazon or Flipkart. It's not a single app or website but a system where many different apps and platforms can connect, giving buyers more options and allowing businesses to sell their products more easily.

Just as UPI gave birth to a new generation of fintech innovators, ONDC is creating fertile ground for specialised technology players who will enrich the ecosystem in unprecedented ways.

Think of it this way - when commerce becomes a shared language (through the Beckn Protocol), it creates opportunities for businesses to specialise in specific 'dialects' of this language. These Specialised Service Providers (SSP) are emerging across various specialised niches:

- Search and Discovery Engines that help buyers find products across the entire network more effectively than traditional e-commerce search
- Inventory Management Systems that help small businesses digitize their catalogs and manage stock across multiple seller apps
- Analytics Providers that give sellers insights about buying patterns across the entire ONDC network, not just a single platform
- Integration Specialists who help existing e-commerce players adapt their systems to speak the 'ONDC language'

Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital

The beauty of this model is that these Specialised Service Providers don't need to build entire marketplaces - they can focus on becoming exceptional at solving specific problems. A small team in Jaipur might build the best AI-powered product categorisation tool, while a startup in Chennai might excel at vernacular search. This specialisation drives innovation much faster than in closed platforms where everything must be built in-house.

This unbundling of commerce technology creates a powerful fly wheel effect:

This proliferation of SSPs will create a virtuous cycle of innovation:



This transformation doesn't just change how people shop; it reimagines the entire commercial landscape, making it more open, inclusive, and dynamic.

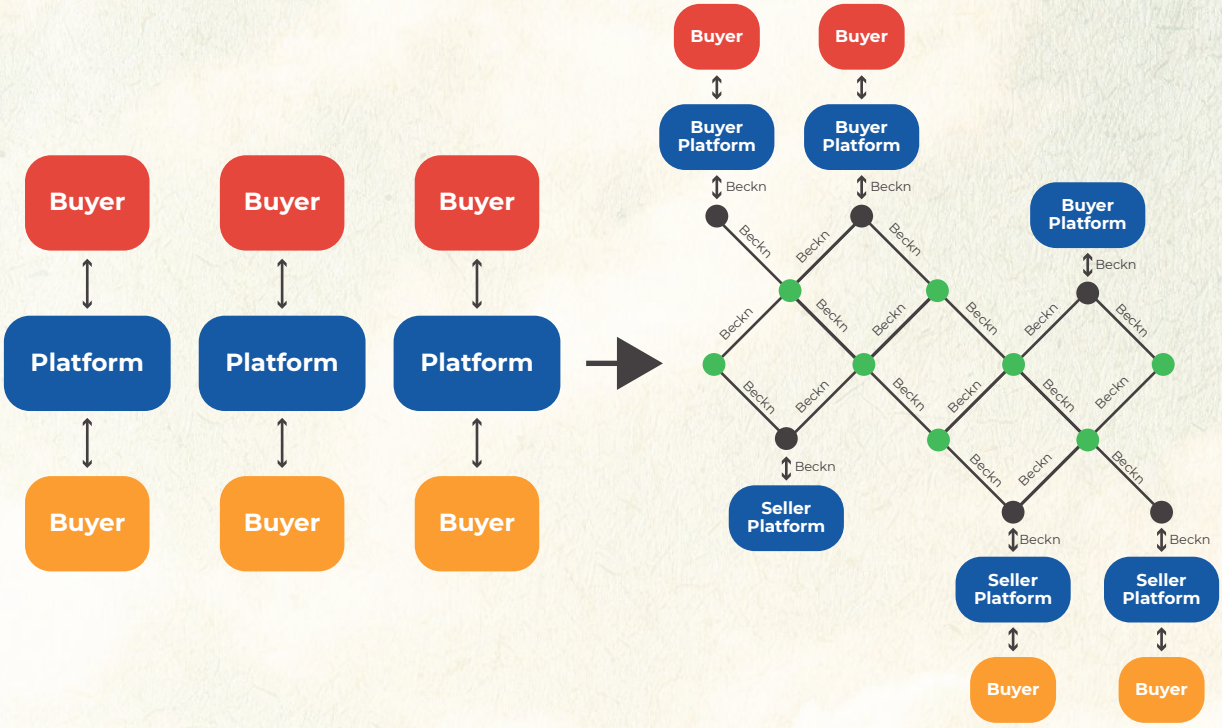
How Does It Work?

The technology that powers ONDC is built on an open protocol called the Beckn Protocol, an open-source protocol that fundamentally reimagines digital commerce. Unlike traditional platforms where sellers must adapt to platform-specific rules, commissions, and operating procedures, Beckn Protocol enables a democratic network where all participating entities (buyers, sellers, logistics providers, and payment services) can interact freely while maintaining their operational autonomy.

Think of Beckn Protocol as a universal language for digital commerce - but not just any language. It's more like a carefully designed system that allows different types of businesses to communicate seamlessly, regardless of what internal systems they use.

Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital



To understand this better, let's start with a real-world analogy. Imagine you're in an international marketplace. Traditionally, if a French seller wanted to trade with a Japanese buyer, they'd need a specific translator who knows both languages. This is similar to how current e-commerce works - each platform has its own "language" (technical specifications), and sellers need to adapt to each platform separately.

Now, imagine if instead, there was a universal translation system that automatically converted everyone's native language in real-time. That's essentially what Beckn Protocol does for digital commerce. It creates a standardized way for different commercial entities to communicate, much like how HTTP standardized how web browsers and servers communicate.

This architecture addresses key pain points of closed ecosystems - from high commission structures and restricted access to customer data, to limited negotiating power and platform-dependent visibility. The protocol's open nature ensures that participants can engage in commerce on their own terms, fostering a more equitable digital marketplace.

This protocol allows different network participants to communicate with each other seamlessly. It's like a common language that all the apps and websites in the ONDC network use to ensure that everything works smoothly together.

Source: Beckn Website

© Copyright 2025 Kalaari Capital

Who Are These Network Participants(NP)?

NPs are different entities on the protocol who fulfil various responsibilities to enable commerce on the network. There are currently 110 NPs live on the network and here are the major NPs that make the system work -

1. **Buyer -** The end consumer who uses ONDC-enabled apps to search for and purchase products or services across various sellers on the network.

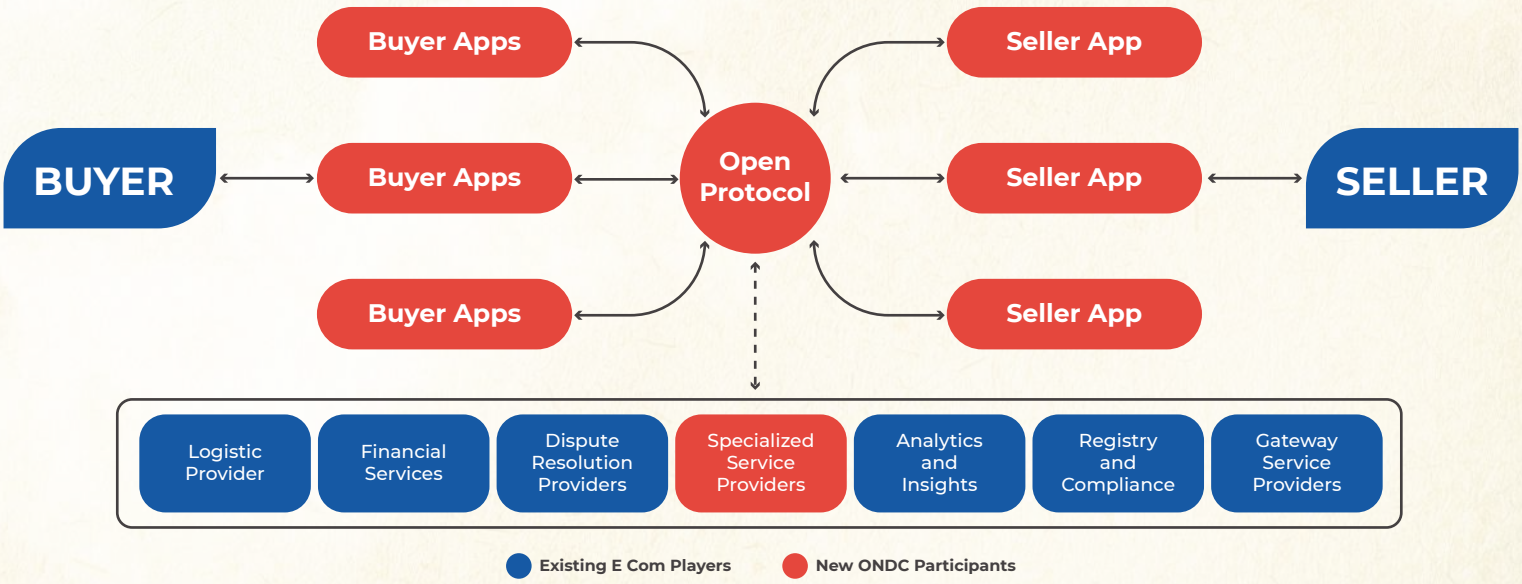
2. **Buyer Apps -** User-facing applications that connect to the ONDC network, allowing buyers to search, compare, and purchase products or services from multiple sellers.

3. **Seller Apps -** Applications that enable sellers to list their products or services on the ONDC network, making them discoverable to buyers across various buyer apps.

4. **Seller -** Any business or individual offering products or services for sale through the ONDC network, ranging from small local shops to large retailers.

5. **Specialized Service Providers -** Entities that create and maintain the technological infrastructure and tools that enable seamless integration and operation of buyer and seller apps on the ONDC network.

ONDC's Open-Network Model



Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital

ONDC IS DESIGNED TO ENABLE ANY "CATALOGUE-ABLE" PRODUCT OR SERVICE

What truly sets ONDC apart is its expansive vision of digital commerce. While most people think of e-commerce as buying physical products, ONDC's architecture supports anything that can be catalogued and transacted - creating entirely new possibilities for digital trade. Think of ONDC as a universal digital marketplace where services, experiences, and products seamlessly converge.

It began with...

- Grocery & FMCG
- Food & Beverage
- Ride Hailing

Now it also has...

- Education, Skilling & Training
- Exports
- Beauty & Personal Care
- Fashion Apparel
- Electronics & Appliances
- Agri Products
- Home & Kitchen
- Health & Wellness
- Metro Ticketing
- B2B

In Early stages...

- Video-on-Demand & Digital Content
- Loan
- Mutual Funds
- Ticketing
- Travel
- Digital Subscriptions
- Insurance

In pipeline

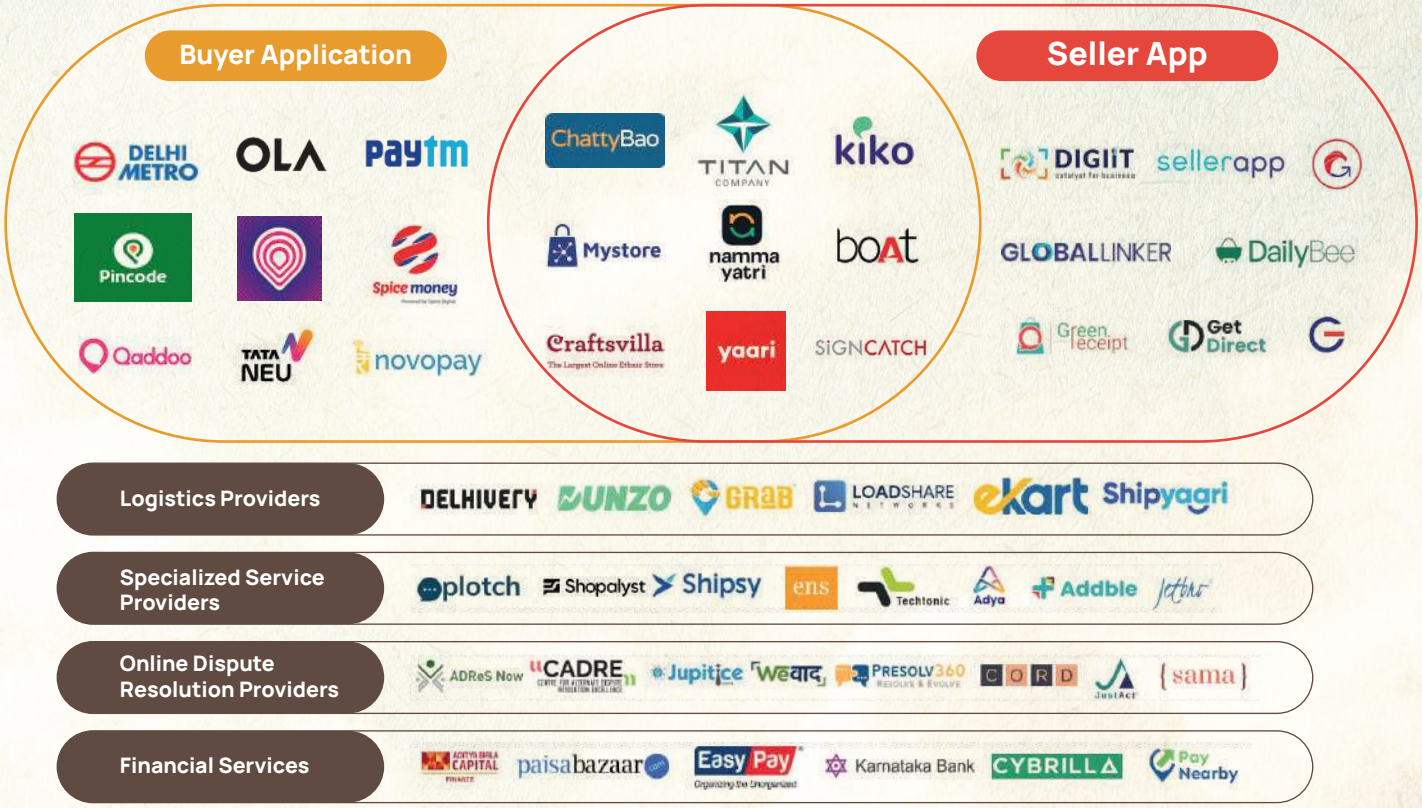
- Repair, Maintenance & At-home services
- Legal Services
- Automotive Supplies
- Sports Equipment
- Warehousing
- Fiber & Yarns

This expanded view of commerce transforms ONDC from just another e-commerce initiative into a comprehensive digital transaction layer for the entire economy. It's not just about selling products anymore - it's about enabling any form of value exchange in the digital space.

Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital

This is how the current map of the market looks like



To access the repository of all the live network particpants on the ONDC network, Click here

Indian Government And Private Sector Entities Have Shown Strong Support For ONDC

The Indian government has played a pivotal role in promoting ONDC, setting up a regulatory framework and providing initial funding to kickstart the project. But various private sector players have shown equal interest in investing in the seed stage of ONDC to promote the grand idea of open commerce. Several notable institutions have acquired stake in ONDC like the BSE, SBI, ICICI, Axis etc.

What Does ONDC Mean For India's Digital Story?

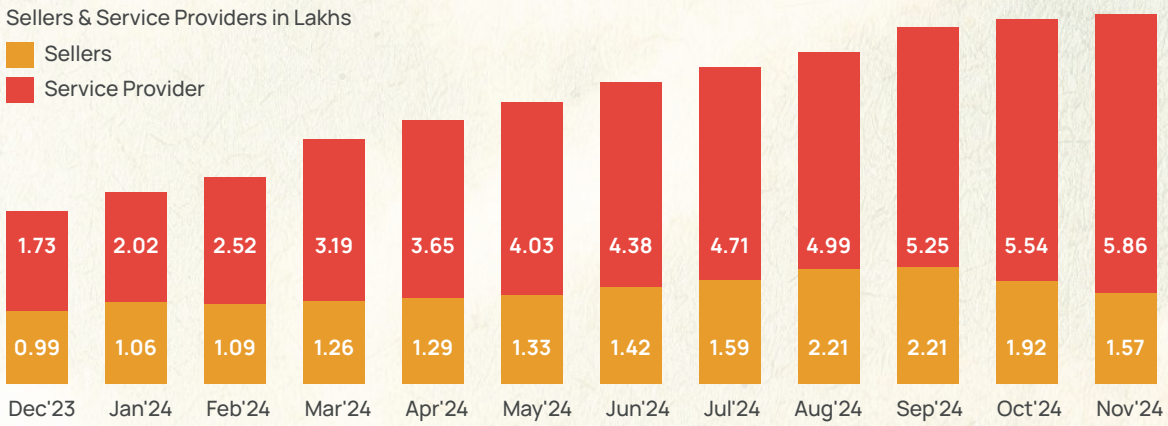
1. Empowering SMEs

One of ONDC's core objectives is to empower SMEs by providing them with nationwide market access. Currently, **India has over 345 million e-retail consumers, yet e-retail penetration is only about 4.3%, which is significantly lower than countries like China (35%) and USA (26%).**

Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital

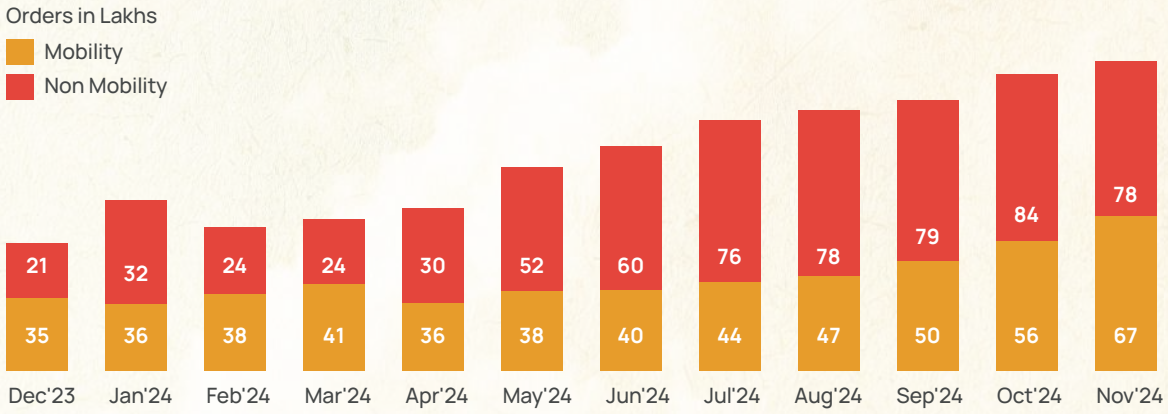
As of August 2024, there are more than 2.2 Lakhs registered sellers who are servicing the customers on the network.



2. Boosting Digital Consumption

ONDC is projected to increase India's digital consumption to **$340 billion by 2030.** This massive growth reflects the network's potential to transform the digital economy by integrating millions of businesses and consumers

ONDC is facilitating more than 1.4 cr orders per month on its network as of November 2024.



3. Inclusivity and Accessibility

ONDC is designed to be inclusive, thus not only enabling easier commerce for consumers and existing sellers but also helping marginalised communities like Kirana stores, Artisans, and small-scale producers, gain a foothold in the digital marketplace. For instance, ONDC plans to make e-commerce accessible to an estimated one million artisans, weavers, and food processors from across India.

As of November 2024, since it's inception, ONDC has facilitated more than 14 cr orders and served more than 806 districts. This is the true power the network holds.

Source: IndiaRetailing, ONDC Website, Moneycontrol, McKinsey

© Copyright 2025 Kalaari Capital

WHAT IS YET TO BE SOLVED FOR? AKA WHITESPACES.

© Copyright 2025 Kalaari Capital

AI DRIVEN CATALOGUING AND INVENTORY MANAGEMENT

Within the evolving ONDC ecosystem, AI-driven cataloging emerges as a critical white space opportunity, offering immense potential for startups and established players alike.

As the network aims to onboard millions of small businesses and create a unified digital marketplace, the need for efficient, accurate, and scalable catalogue and inventory management becomes paramount. Startups entering this space have the chance to become foundational pillars of the ONDC infrastructure, providing essential services that can shape the future of digital commerce in India.

It's important to note that the opportunities in this space are not yet comprehensive or fully realised. As ONDC evolves and matures, new challenges and needs will emerge, creating additional avenues for innovation and entrepreneurship in AI-driven cataloging solutions.

Why AI-Driven Cataloging and Inventory Management Matters?

Imagine you go over to your favourite buyer app to buy your monthly groceries to be delivered home only to know post checkout that few of the items you ordered are out of stock. **This is poor Customer experience (CX) 101.**

The Cataloguing Challenge

In the ONDC ecosystem, AI-driven cataloging addresses the above and several other critical needs:

1. **Standardisation :** ONDC aims to create a standardised format for product listings across diverse sellers. AI can help in automatically translating varied catalog formats into a unified ONDC-compliant structure.

2. **Scale :** With millions of SKUs from numerous sellers, manual catalog management becomes unfeasible. AI solutions can handle this scale efficiently.

Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital

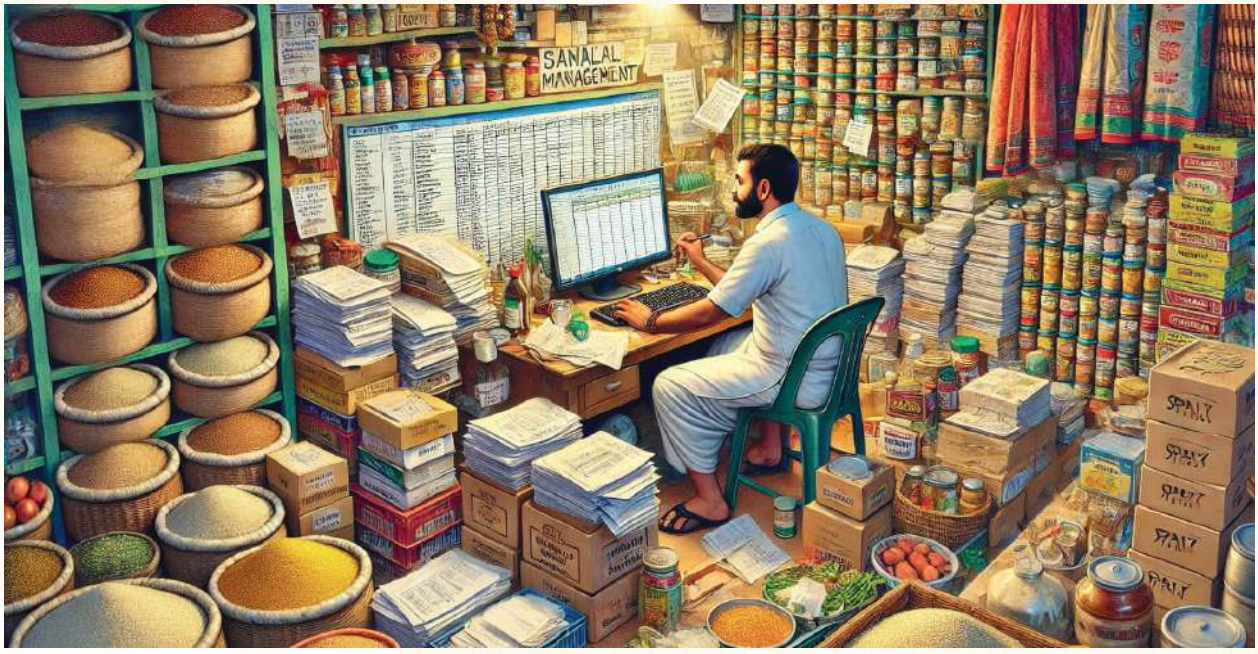
3. **Local Language Support :** India's linguistic diversity necessitates multilingual catalog support. AI-powered translation and localization can make products discoverable across language barriers.

4. **Trust and Accuracy:** By reducing errors and inconsistencies in product listings, AI cataloging can build trust among buyers on the open network.

Traditional catalog creation requires extensive manual effort to photograph, describe, and categorize each product. For a typical Kirana store with 2,000-3,000 SKUs, this process could take weeks. Additionally, maintaining consistency across product descriptions, categories, and pricing across multiple seller apps becomes a logistical nightmare.

The Inventory Management Pain Point

Parallel to cataloguing, inventory management poses its own set of challenges: Traditional e-commerce inventory management involves a significant amount of manual work, including categorizing products, updating listings, and managing inventory. This manual process is time-consuming and prone to errors, which can lead to incorrect data entry, misclassification, and outdated information, ultimately affecting customer experience and sales.



Dall-Es interpretation of an Indian Kirana store owner having to sell and manage inventory at the same time

Costs associated with manual catalog management are substantial. Businesses need to hire additional staff, invest in training, and manage the workforce to keep catalogs updated, which can be financially straining, especially for smaller companies

Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital

What's there to be built?



1. AI-Powered Catalog Creation Tools:
Develop user-friendly tools that allow small businesses to create professional-grade catalogs using AI and computer vision.



2. Multilingual Catalog Management:
Create solutions that automatically translate and localize product information across Indian languages.



3. AI-Driven Product Matching:
Develop algorithms to match similar products across sellers, enhancing comparison shopping on ONDC.



4. Semantic Search Enhancements:
Build AI models that understand context and intent in search queries, improving product discovery.



5. Real-Time Stock Monitoring:
Computer vision systems that can track shelf inventory through regular smartphone photos or basic CCTV cameras.



6. Predictive Stock Management:
AI algorithms that analyze historical sales data to predict demand and suggest optimal stock levels.



7. Automated Reordering Systems:
Smart systems that can automatically generate purchase orders based on inventory levels and predicted demand.

During our conversation with Girish Ramachandra, founder of Shopalyst (an SSP on ONDC and a Kalaari portfolio company), he highlighted a critical need: universal catalog synchronization. Currently, sellers managing catalogs across multiple buyer interfaces face inconsistencies and poor customer experience. The solution lies in a "create once, publish everywhere" system that would:



Auto-sync product information across buyer interfaces



Enable real-time inventory updates network-wide



Maintain consistency in product details



Handle format conversions for different buyer apps

Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital

Potential Impact

Enhanced Customer Experience: AI-driven cataloguing and inventory solutions together create a seamless shopping experience, with accurate product information and real-time stock visibility. This dual enhancement leads to reduced order cancellations, better product discovery, and increased customer satisfaction, resulting in higher conversion rates and repeat purchases.

Operational Efficiency: The combination of automated cataloguing and smart inventory management significantly reduces manual workload. While cataloguing automation handles product listings and updates, inventory systems ensure optimal stock levels, allowing teams to focus on business growth rather than routine tasks.

Market Competitiveness: Businesses leveraging both AI-driven cataloguing and inventory management gain a strong market advantage. They can offer accurate product information and maintain optimal stock levels while reducing inventory costs. This enables them to compete effectively with larger players by providing better availability, pricing, and product information across all seller apps.

Economic Impact: The integration of smart cataloguing and inventory solutions democratizes digital commerce for millions of MSMEs. By providing enterprise-grade tools at accessible costs, these technologies enable small businesses to compete effectively in the digital marketplace. This technological democratization drives faster business scaling and broader digital adoption, contributing significantly to economic growth across urban and rural India.

As ONDC grows, the demand for sophisticated AI cataloging and Inventory management solutions will increase, offering continuous opportunities for innovation and entrepreneurship in this space.

In conclusion, this represents a critical and evolving opportunity within the ONDC ecosystem. By addressing core challenges in product information management, discovery, and standardization, startups in this space can play a pivotal role in shaping the future of India's digital commerce landscape.

Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital

MULTILINGUAL AND MULTI MODAL CONSUMER INTERFACES

As ONDC seeks to democratize digital commerce across India's linguistically diverse landscape, multilingual support and Multi Modal commerce emerge as critical enablers. This whitespace addresses the challenge of language barriers that often hinder e-commerce adoption, especially in tier 2 and 3 cities and rural areas. By integrating vernacular language support and AI-powered conversational interfaces, ONDC can significantly expand its reach and inclusivity. This approach not only makes digital commerce more accessible to non-English speaking populations but also enhances user engagement and trust through personalized, language-specific interactions.



If we want to expand our digital commerce landscape beyond the current 345 million e-retail consumers, the need for intuitive, language-native, and voice-enabled interfaces becomes critical.

Startups entering this space have the opportunity to become essential enablers of ONDC's vision, creating solutions that can help the network reach India's next billion users.

Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital

Why Multilingual And Multi-Modal Interfaces Matter?

Imagine a small trader in rural Rajasthan trying to list his traditional handicrafts on ONDC, or a homemaker in Tamil Nadu searching for specific grocery items. Without interfaces in their preferred languages and input methods, they're effectively excluded from the digital commerce revolution.

In the ONDC ecosystem, multilingual and multi-modal interfaces address several critical needs:

Language Coverage:
India has 22 official languages and 121 spoken languages with over 10,000 speakers. Yet, only ~10% of India's population can speak English, while digital platforms predominantly operate in English.

Flipkart sees **1.5 million** daily users in local languages, with vernacular user share growing from **12% to 18%** in one year

Amazon India witnessed a **3x surge** in new customers within five months of launching Hindi support

Meesho achieved a **20%** adoption rate after introducing Hindi, with 80% of their user base coming from tier 2 cities and saw an **5.5 x increase** in transacting user base

Voice First Approach:
70% of the worlds internet users preferring voice search over typing, voice-enabled commerce becomes not just an alternative but a necessity.

Digital Literacy Bridge:
Approximately 311 million Indians cannot read a newspaper or simple texts, multi-modal interfaces can serve as a crucial bridge to digital commerce.

72% of consumers are more likely to buy a product or service when the information is presented in their native language

Source: Indianexpress, TheHistoryOfEnglish, Process9, Noida Business Guide

© Copyright 2025 Kalaari Capital

What's There To Be Built?

Voice-First Commerce Platforms

- Develop voice-enabled buyer apps that support natural language interactions across Indian languages
- Create voice-based catalog management tools for sellers
- Enable voice-driven search and discovery across the ONDC network

Vernacular Search Engines

- Build search algorithms that understand context in regional languages
- Handle code-mixed queries effectively. Think Hindi mixed with English or Tamil mixed with English
- Support phonetic search for product discovery

Multi-modal Input Systems

- Create hybrid interfaces combining voice, text, and visual inputs
- Develop gesture-based interfaces for specific user segments
- Enable seamless switching between different input modes

Language Translation Infrastructure

- Real-time catalog translation across Indian languages
- Context-aware translation considering regional commerce terminology
- Automated content localization for product descriptions

The Potential Impact

Revolutionizing E-Commerce Penetration in Underserved Markets

India's e-commerce penetration is currently skewed towards metro cities, with rural areas and Tier 2 and Tier 3 cities remaining largely untapped. Multilingual and multimodal consumer interfaces can change this dynamic by making digital commerce more accessible to underserved regions.

E-commerce penetration in rural and semi-urban areas is less compared to its metro counter-parts. By enabling regional-language and multimodal access, ONDC could drive a surge in **increase of e-commerce adoption** in these regions, adding millions of new users to the digital economy.

Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital

Empowering Non-English Speakers

Approximately 73% of internet users in India prefer Indian languages, forming a robust regional language user base of 540 million people and unlocking a $53 billion market opportunity.(Standard, 2024). Current e-commerce platforms that prioritize English alienate these users.

Non-English speaking buyers can shop in their native language, making the digital marketplace more relatable and trustworthy. Small and local sellers from regional markets can list products and engage with buyers in their native language, expanding their reach.

Enhancing Digital Literacy and User Accessibility

Multimodal interfaces (combining voice, visual, and text-based interactions) can drastically reduce the barriers posed by low literacy levels and limited technical proficiency.

Voice-based navigation can help users who are uncomfortable with typing or reading text-heavy interfaces. For instance, a shopper can say “buy rice in Kannada” to browse products in their language without needing to navigate a text-based platform. Simple, image-driven navigation, such as clicking on product icons, allows even semi-literate users to shop confidently.

Driving Competition and Innovation

Startups and tech companies that create multilingual and multimodal solutions for ONDC can differentiate themselves in the marketplace, driving innovation and competition. Platforms that provide superior user accessibility will stand out in the crowded e-commerce ecosystem.

Platforms with accurate regional language support and intuitive multimodal interfaces can increase conversion rates. As users are more likely to complete purchases when they feel understood and supported. Buyers and sellers on these platforms are likely to stay longer and engage more, creating a virtuous cycle of growth and loyalty.

Source: Creative-words

© Copyright 2025 Kalaari Capital

INTEGRATED VALUE CHAINS

The concept of Integrated Value Chains holds transformative potential within the ONDC framework. By digitizing each step in a product's lifecycle—ranging from sourcing and manufacturing to logistics, retail, and post-sale services—ONDC enables small and large businesses alike to participate in a seamlessly connected digital commerce ecosystem. This whitespace offers **massive opportunities for startups and founders** to build solutions that tackle inefficiencies by collaboration, and leverage the rich data generated by these chains to create entirely new markets and businesses.

Why Integrated Value Chains Matter?

India's MSME sector, comprising over 63 million enterprises and contributing approximately 30% to India's GDP, still operates largely through fragmented, manual processes. The current digital transformation wave has primarily focused on the front-end customer experience, leaving vast opportunities in backend operations untapped.

Consider these stats:

Digitising supply chains allows companies to reduce their inventory by

10 to 20%

India loses

$65 billion

every year due to inefficient supply chain systems

The opportunity here, lies in creating specialized layers within ONDC that can digitize and integrate each of these steps seamlessly. This isn't just about building another B2B marketplace – it's about creating digital highways that connect every participant in the value chain through the open protocol architecture of ONDC.

Source: Moneylife, MSME Ministry 2023

© Copyright 2025 Kalaari Capital

Here's How The Digitization Of Each Step Could Unfold:



Raw Material Sourcing
A specialized interface within ONDC where manufacturers can discover and directly connect with multiple raw material suppliers. The system could integrate real-time pricing, quality certifications, and availability status – similar to how UPI made financial transactions transparent and immediate.



Production Planning
Digital tools could help manufacturers optimize their production schedules based on real-time demand signals from across the ONDC network. This would be particularly transformative for MSMEs who often struggle with demand forecasting.



Inventory and Warehouse Management
Smart inventory systems could automatically sync with both upstream suppliers and downstream buyers, creating a fluid, real-time view of stock levels across the entire value chain. This addresses a critical pain point highlighted in the white paper – the current disconnect between digital commerce and physical inventory management.



Quality Control and Certification
Digital quality verification systems could be integrated into the chain, allowing for transparent tracking of certifications and compliance – crucial for sectors like pharmaceuticals or food processing where trust and verification are paramount.

The Power Of This Integration Lies In Its Network Effects.

Just as how UPI's transactions grew 100x in six years, digitized value chains could create similar exponential growth in B2B commerce efficiency. The key is maintaining ONDC's core principles of openness and interoperability while addressing industry-specific needs.

The digitization of these value chains through ONDC creates an unprecedented data opportunity too. Every transaction, every movement, and every interaction generates valuable data points that can be leveraged for insights.

Unlike traditional e-commerce platforms where data remains siloed, ONDC's structure creates a rich tapestry of interconnected commercial activities.

Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital

This Opens Up Several Sophisticated Data Plays:



Commerce Intelligence Layer

An intelligence layer that analyzes transaction patterns across the entire network. This could help predict market trends before they become apparent, similar to how weather forecasting evolved from local observations to sophisticated predictive models. For instance, a surge in demand for certain ingredients in Kerala could indicate an emerging food trend that businesses in Punjab could prepare for.



Price Optimization Engines

The network's transparency in pricing creates opportunities for sophisticated arbitrage detection and price optimization tools. These wouldn't just compare prices but could understand price elasticity across different markets, seasons, and consumer segments. A vegetable vendor in Nashik could optimize their pricing based on real-time demand patterns across multiple cities.



Supply Chain Intelligence

The integration of multiple players in the value chain creates opportunities for predictive supply chain analytics. Think of it as a Bloomberg terminal for local commerce – providing real-time insights into inventory levels, supply bottlenecks, and demand forecasts. This could revolutionize how small businesses manage their operations.



Hyperlocal Market Intelligence

By analyzing transaction patterns at a granular level, businesses could understand micro-market dynamics. A small retailer in Jaipur could understand not just what sells in their neighborhood, but why it sells and how the pattern differs from similar neighborhoods in other cities.



Risk Assessment Models

The rich transaction data could enable new approaches to business risk assessment. Financial institutions could develop more accurate lending models based on actual business performance rather than traditional metrics. This could be particularly transformative for India's vast MSME sector.

The potential impact on India's economy could be substantial. By making data-driven decision-making accessible to millions of small businesses, these data plays could help unlock the efficiency and growth potential of India's vast informal sector.

Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital

CONCLUSION

India's digital transformation has always been characterized by ambitious vision meeting grassroots innovation. From Aadhaar to UPI, each milestone has built upon the last. We hope ONDC stands as the most transformative chapter in this journey yet.

ONDC's compelling power lies in addressing the fundamental asymmetries in India's digital commerce landscape. While e-commerce has flourished in urban India, it has remained largely an urban phenomenon. ONDC's architecture – open, inclusive, and interoperable – promises to change this narrative.

The white spaces we've explored – from AI-driven cataloging to multilingual interfaces and integrated value chains – form the building blocks of an inclusive digital economy. When a small kirana store can efficiently manage inventory through AI tools, list products in local languages, and plug into national supply chains, it transforms itself and the ecosystem.

At **Kalaari Capital**, we see ONDC as a catalyst for India's next wave of entrepreneurial innovation. We're particularly excited about entrepreneurs who think deeply about these white spaces and build solutions that can scale with ONDC's growing network. As investors deeply committed to India's technology ecosystem, we're actively looking to partner with founders who share this vision and are building solutions that can accelerate ONDC's adoption and impact.

Source: Kalaari Analysis

© Copyright 2025 Kalaari Capital

THANK YOU,

For further information, please contact

Vani Kola
vani_kola@kalaari.com

Gautam Narayan
gautam@kalaari.com

About Kalaari
Kalaari Capital is an early-stage, technology-focused venture capital firm based in Bengaluru, India. Since 2006, Kalaari has empowered visionary entrepreneurs building unique solutions that reshape the way Indians live, work, consume and transact. The firm's ethos is to partner early with founders and work with them to navigate the inevitable challenges of fostering ideas into successful businesses. At its core, Kalaari believes in building long-term relationships based on trust, transparency, authenticity, and respect.

This Report has been prepared by Kalaari Capital and is intended for informational purposes only. This report is prepared based on Kalaari Capital's own research, internal data and on the basis of publicly available information, and other sources believed to be reliable, but which may not have been verified independently. Kalaari Capital makes no representations or warranties, express or implied, for the accuracy, completeness, reliability, correctness, or use of any information in this Report and shall not be responsible and/or liable for any error, omission or inaccuracy of such information. This Report is not intended to constitute investment advice or any other type of professional advice or recommendation of any particular transaction or investment, nor does it constitute an offer to sell, or solicitation of an offer to buy, any securities or other financial instruments. Recipients of this Report are encouraged to seek independent advice tailored to their specific circumstances before making any decisions based on the content of this Report.

Kalaari Capital retains exclusive ownership of the intellectual property rights to the Report. This Report, or any part of this Report, cannot be copied, reproduced, distributed, or transmitted in any form or by any means without the prior written consent of Kalaari Capital. Kalaari Capital expressly disclaims any and all liability for any losses, damages, or costs arising from the use of or reliance on the information presented in this Report. By accessing this Report, you acknowledge and agree to the terms set forth in this disclaimer.

© Kalaari Capital Advisors Private Limited, 2025